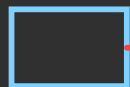
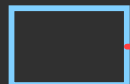
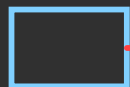


00 MOUNTING OVERVIEW

Three retroreflective beam axes; passive reflector carrier is optional FR-4.

WALL A - SENSOR / CONTROLLER CARRIER

50 x 240 mm, electronics + removable sensors



BEAM 3 - +200 mm

BEAM 2 - +100 mm

BEAM 1 - DATUM 0 mm

WALL B - PASSIVE REFLECTORS

30 x 240 mm carrier; no electrical PCB required



100 mm

100 mm

Align by sensor optical axes - not by carrier edges. Use slots/shims for pitch and yaw.

Reflectors need only a rigid mounting plate; blank FR-4 is one convenient option.

Title: **MOUNTING OVERVIEW**

Project: **GarageBeamSafety.PrjPcb**

Engineer: **Henrique Rosa**

Date: 01/09/2026 Sheet 01 of 04

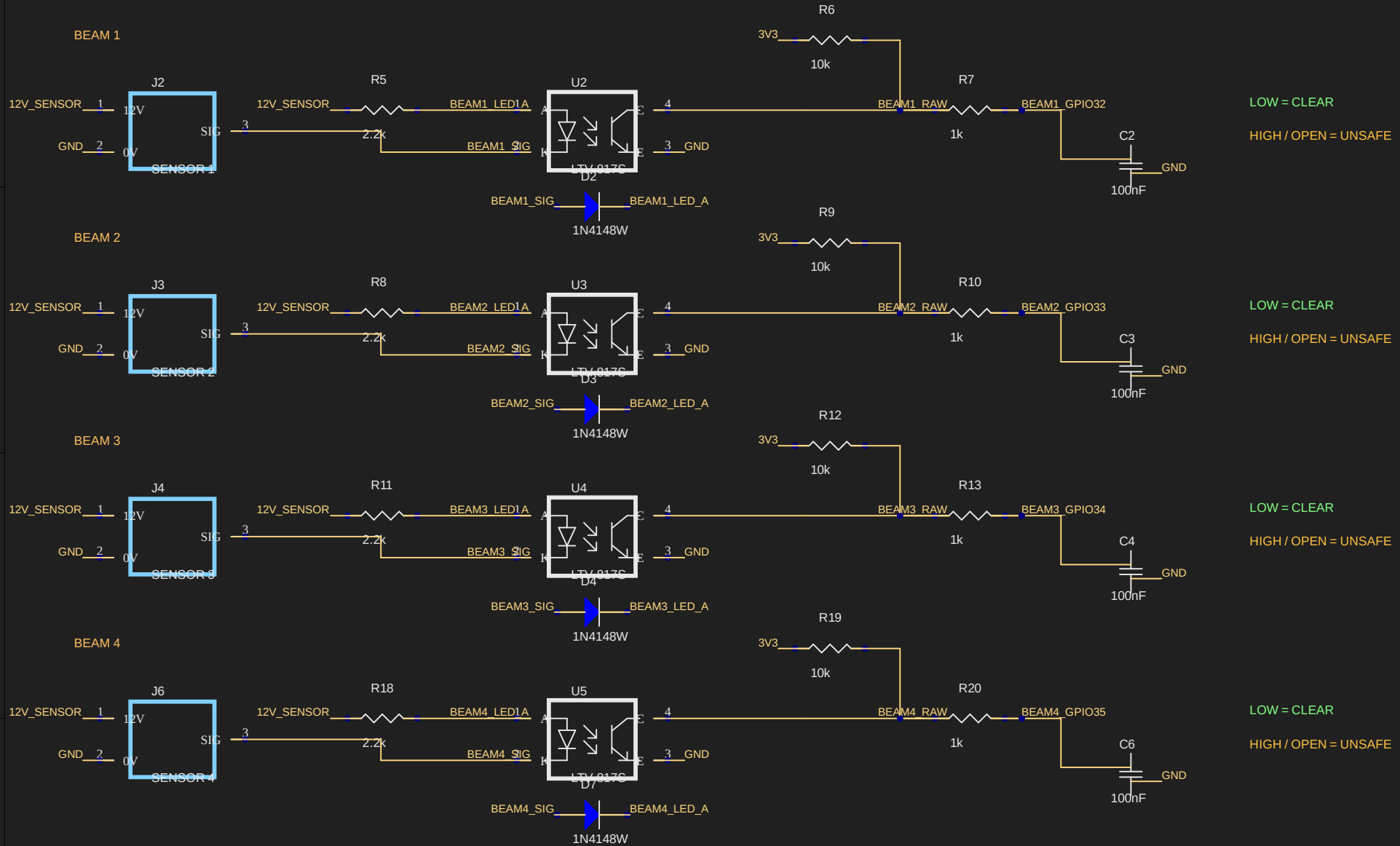
version: A

File: 00_Mounting_Overview.SchDoc



02 FOUR CURRENT-PROVING SENSOR INPUTS

NPN Light-ON sensors: current must flow for CLEAR; open or unpowered channels are unsafe.



Enable only installed beams. LOW for 500 ms earns CLEAR. Signal-to-GND shorts / stuck-ON sensors are NOT detected.

Title: **BEAM INPUTS**

Project: **GarageBeamSafety.PrjPcb**

Engineer: **Henrique Rosa**

Date: **02/09/2026**

Sheet **03**

of **04**

version: **A**

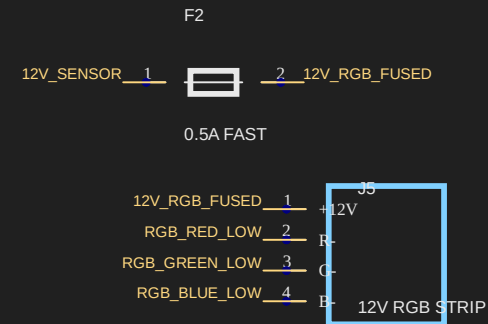
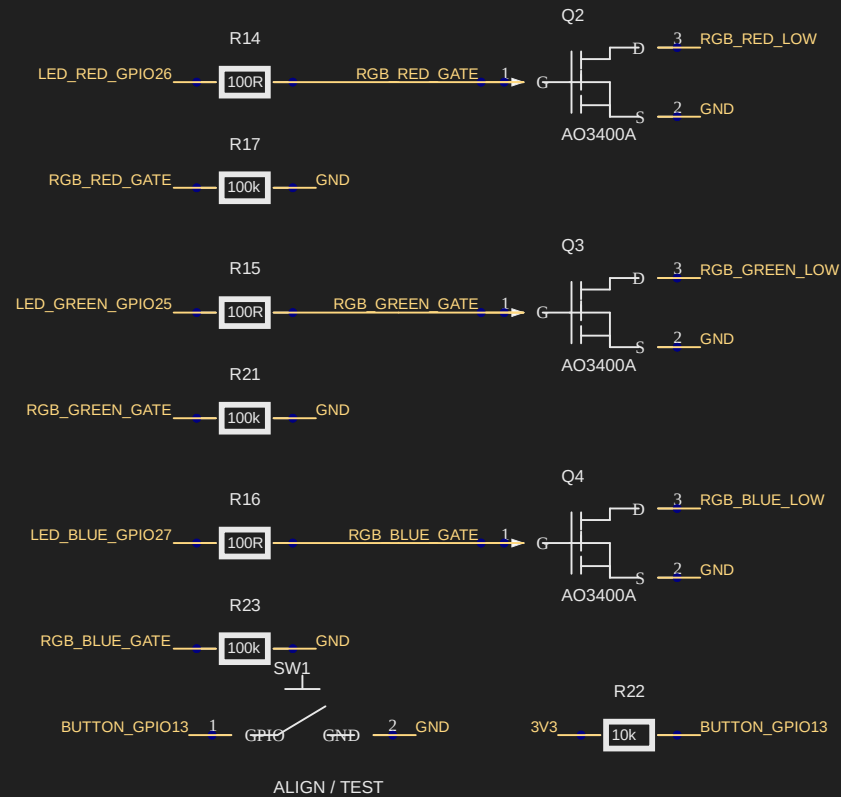
File: **02_Beam_Inputs.SchDoc**



03 VISIBLE RGB STATUS - NO BUZZER

Steady green=CLEAR; steady red=BLOCKED; flashing amber=BOOT / FAULT / UNKNOWN. Dark never means clear.

3.3V GATE DRIVE / 12V LED LOAD



J5 PINOUT CHANGED: 1=+12V 2=R- 3=G- 4=B-

15cm of 12V 5050 RGB strip, 60 LED/m; common anode.

Strip MUST include its own current limiting. No bare power LEDs.

Short local cable; 250mA/channel maximum design load.

GPIO14 is spare. Lamp current/failure sensing is NOT provided.

ALIGNMENT / TEST BUTTON ON LEFT

ADVISORY ONLY - no relay, motor, or garage-door actuation connection is provided.

Title: *VISIBLE RGB STATUS*

Project: *GarageBeamSafety.PrjPcb*

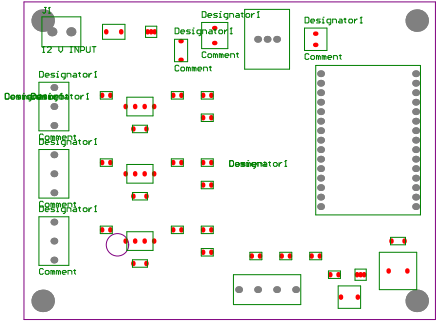
Date: 06/09/2026 Sheet 04 of 04

File: 03_UI_Outputs.SchDoc

Engineer: Henrique Rosa

version: A





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D3 PASSIVE REFLECTOR CARRIER - NO POWER / NO COPPER



REFLECTOR 3 / +200 mm



REFLECTOR 2 / +100 mm



REFLECTOR 1 / 0 mm



Board Stack Report